

Is Habitat Protection Working?

Is Habitat protection working?

Natural area protection is one of the key management strategies for preserving biodiversity. However, many landscapes are under strong economic and development pressures, and the effectiveness of conservation measures must be demonstrated.

A few years ago, the government implemented protection measures across several natural sites, including wetlands, forests, and grasslands. These measures included limiting human access (e.g., restricting vehicles and public entry), restoring vegetation, and controlling invasive species.

Data were collected: **Before protection (baseline)** and **After 2 years of protection**. At each site, local managers measured:

- 🌱 **Species richness** (number of species)
- 🌿 **Vegetation cover (%)**
- 🐸 **Abundance of an indicator species** *European tree frog (Hyla arborea)* (Sensitive to environmental change and Reflect ecosystem health)

A preliminary report compared the “before” values vs the “after” values , aiming to demonstrate that the protection has improved the natural areas. However, the conclusion was ⚠️ **“There is no clear improvement after protection.”**

And the government is facing the decision to revert area protection. A key aspect that the preliminary study didn’t consider is that the protected areas are very different from each other. Some are naturally rich ecosystems, other are degraded, some are large, others small. This creates a huge between-site variability.

Your mission

1. Reanalyse the data properly
2. Decide: Is protection working?
3. Communicate: A recommendation to policymakers

The Data

```
set.seed(42)
n_sites <- 20
site <- paste0("Site_", 1:n_sites)
habitat <- sample(c("Wetland", "Grassland", "Forest"), n_sites, replace = TRUE)

# --- Species richness (counts) ---
richness_before <- pmax(0, round(rnorm(n_sites, 20, 10)))
```

```

# --- Vegetation cover (0-100%) ---
veg_before <- pmin(100, pmax(0, rnorm(n_sites, 50, 20)))

# --- Frog abundance (counts) ---
frog_before <- pmax(0, round(rnorm(n_sites, 8, 4)))

# --- Improvements ---
richness_after <- pmax(0, round(richness_before + rnorm(n_sites, 3, 2)))
veg_after <- pmin(100, pmax(0, veg_before + rnorm(n_sites, 8, 5)))
frog_after <- pmax(0, round(frog_before + rnorm(n_sites, 2, 1.5)))

# --- Introduce declines ---
decline_sites <- sample(1:n_sites, 3)
richness_after[decline_sites] <- pmax(0, round(richness_before[decline_sites] - rnorm(3, 2, 1)) )
veg_after[decline_sites] <- pmin(100, pmax(0, veg_before[decline_sites] - rnorm(3, 5, 2)) )
frog_after[decline_sites] <- pmax(0, round(frog_before[decline_sites] - rnorm(3, 2, 1)))

# --- Final dataset ---
data <- data.frame( site, habitat, richness_before, richness_after, veg_before, veg_after, frog_befc

data

```

	site	habitat	richness_before	richness_after	veg_before	veg_after
1	Site_1	Wetland	20	25	32.40327	43.734313
2	Site_2	Wetland	33	39	68.93826	81.025984
3	Site_3	Wetland	30	27	35.90897	31.081187
4	Site_4	Wetland	29	31	28.43661	40.367349
5	Site_5	Grassland	25	27	48.96674	51.313204
6	Site_6	Grassland	30	35	61.61858	61.542385
7	Site_7	Forest	15	18	15.00000	18.000000
8	Site_8	Wetland	23	22	0.00000	0.000000
9	Site_9	Forest	18	21	29.93791	38.152430
10	Site_10	Forest	39	40	57.47108	64.751661
11	Site_11	Wetland	37	38	53.20444	60.479047
12	Site_12	Wetland	24	26	23.18250	30.989289
13	Site_13	Grassland	23	24	39.81100	44.589490
14	Site_14	Grassland	16	19	0.00000	9.924926
15	Site_15	Grassland	28	31	79.97552	80.241272
16	Site_16	Forest	27	25	62.50487	57.609805
17	Site_17	Forest	11	14	50.75534	56.873889
18	Site_18	Wetland	20	24	56.06547	70.271049
19	Site_19	Wetland	41	46	34.36343	37.624754
20	Site_20	Forest	22	24	44.35861	61.198495

	frog_before	frog_after
1	15	18
2	10	11
3	0	0
4	12	13
5	8	7
6	8	11
7	7	11

8	12	11
9	7	9
10	9	10
11	7	9
12	15	15
13	10	14
14	9	10
15	14	16
16	9	7
17	3	5
18	9	11
19	2	2
20	11	15



THINK FIRST (before any analysis)



Question 1. Expectation

Do you expect all sites to improve after protection?

- Yes / No
- Why?



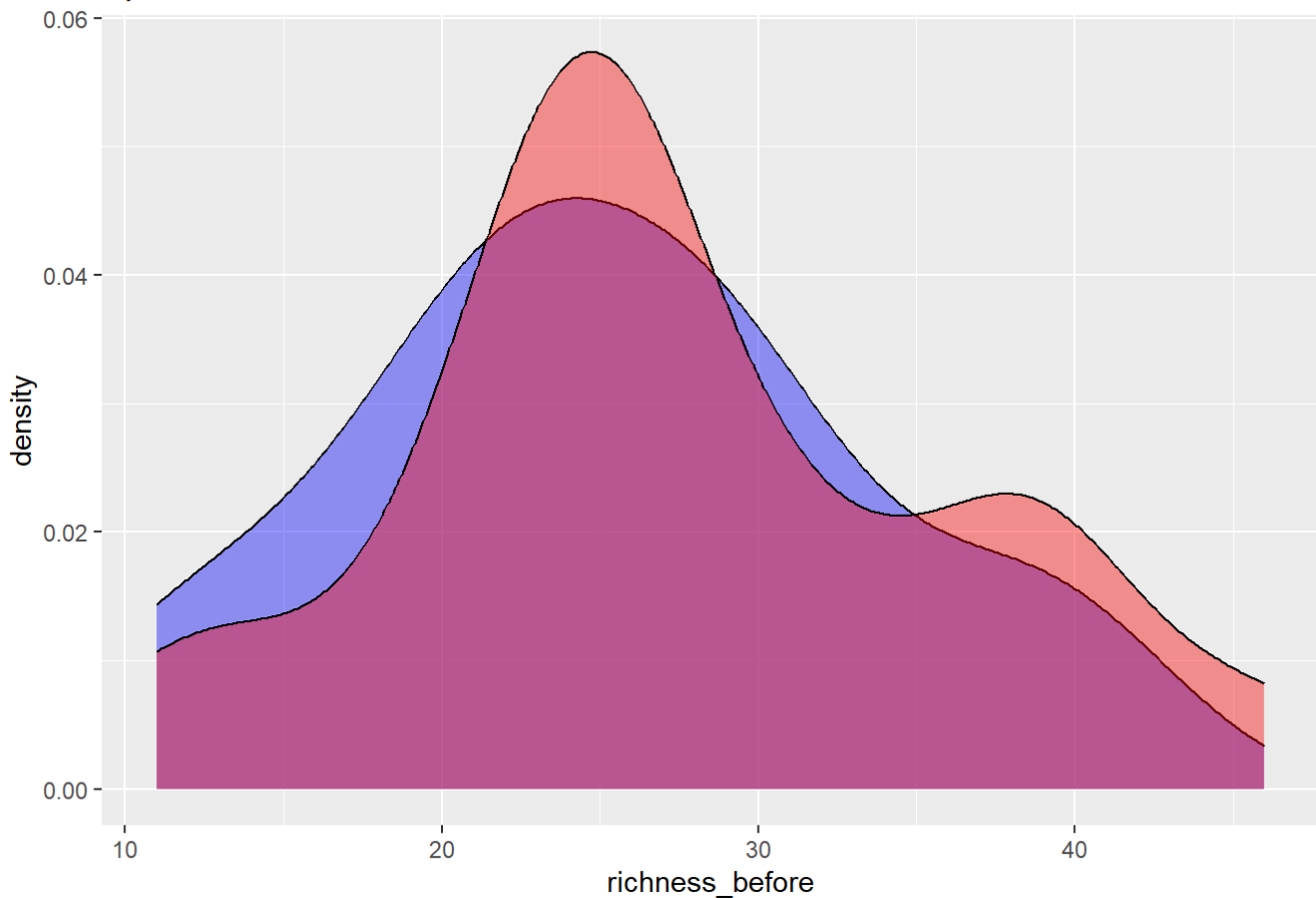
Visualisation 1

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.5.3

```
ggplot(data) +  
  geom_density(aes(x = richness_before), fill = "blue", alpha = 0.4) +  
  geom_density(aes(x = richness_after), fill = "red", alpha = 0.4) +  
  labs(title = "Species Richness: Before vs After")
```

Species Richness: Before vs After



Question 2. First observations

Based on this plot:

 Does protection seem effective?

- Yes / No / Unclear
- Why?

Can you think (and draw/code) an alternative visualisation for this data? When facing this questions you can always explore <https://datavizcatalogue.com/> or <https://r-graph-gallery.com/>

Should we check if this data is normally distributed?

►  Click for discussion

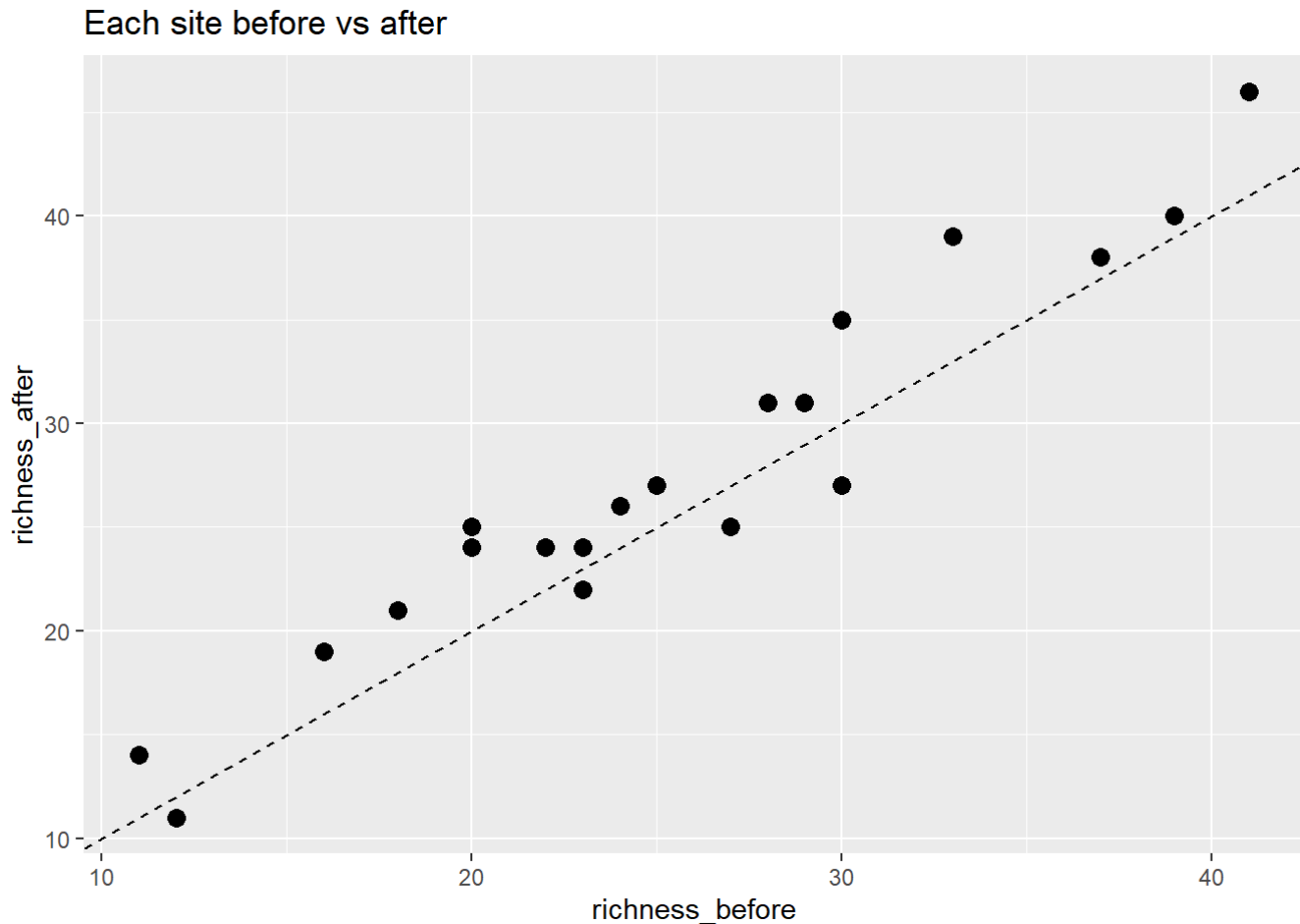
Why might this be misleading?

Tip

Sites are very different from each other!

Visualisation 2: Pairing

```
ggplot(data, aes(x = richness_before, y = richness_after)) +  
  geom_point(size = 3) +  
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +  
  labs(title = "Each site before vs after")
```



Question 3. Scatter plot interpretation

What does it mean if a point is:

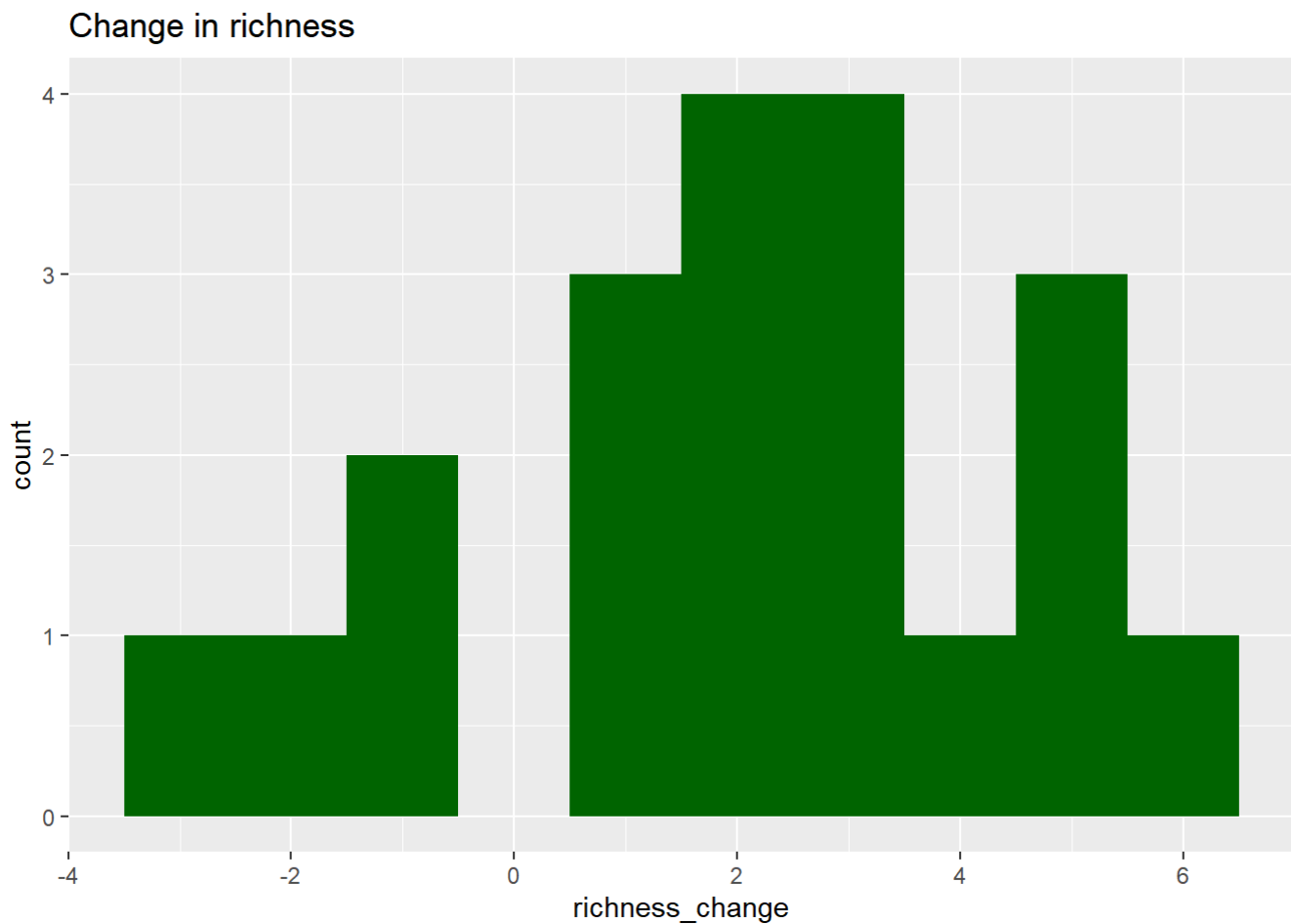
- Above the line?
- Below the line?

►  Answer

Visualisation 3: CHANGE

```
data$richness_change <- data$richness_after - data$richness_before  
  
ggplot(data, aes(x = richness_change)) +
```

```
geom_histogram(bins = 10, fill = "darkgreen") +  
labs(title = "Change in richness")
```



👉 Question 4. Richness change

Now answer:

- 👉 Is the **average change** positive? (Calculate it in R, you know how).
- 👉 Are the differences (after-before) approximately normally distributed?
- 👉 Are all sites improving?

Can you add a vertical line at the 0 value so we can clearly distinguish the sites that improved and the sites that went worse? Try to add as well the average value.

► 💡 Click for discussion

🧠 Concept Check

Warning

✗ Wrong question
Are before and after different?

✓ Correct question

Is the **change within sites** different from zero?



Statistical Tests



Paired t-test

```
t.test(data$richness_after, data$richness_before, paired = TRUE)
```

Paired t-test

```
data: data$richness_after and data$richness_before
t = 3.7601, df = 19, p-value = 0.001325
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 0.9088952 3.1911048
sample estimates:
mean difference
      2.05
```



Ignoring pairing

```
t.test(data$richness_after, data$richness_before, paired = FALSE)
```

Welch Two Sample t-test

```
data: data$richness_after and data$richness_before
t = 0.75905, df = 37.851, p-value = 0.4525
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
-3.418102  7.518102
sample estimates:
mean of x mean of y
 27.45    25.40
```



Question 5. Compare tests

Compare the two results:

- Which test detects an effect?
- Why are they different?

Indicator Species: Frog

Now it is your turn, provide further evidence for the impacts of nature protection with the data of the indicator species. This time you will have to code it yourself.

```
# Calculate frog_change here>

# Visualise the distribution of frog_change>
```

Question 6. Frog abundance

Is the frog population responding to protection?

Why might frogs be especially informative?

Question 7 (open interpretation)

Why might some sites decline?

Think biologically: - Habitat quality? - Time lag? - External disturbances?

FINAL TASK (Policy Report)

You must advise the government.

Your Report

Answer:

1. Is habitat protection working?
 2. Is biodiversity increasing?
 3. Is vegetation improving?
 4. Is the frog population responding?
 5. Should protection continue?
-

Test yourself

Important

1. Pairing removes baseline differences between sites, but how could we account for broader environmental changes, such as global temperature increases?
2. Even when data are naturally paired, should we always use a paired test? Can you think of an example where, despite having a before/after design or repeated measurements, a paired test might not be appropriate?
3. Since not all sites improved, would you suggest further data exploration? For example, should we analyse results by habitat type? What other factors might determine the success of area protection, and therefore, what additional variables would you include in the dataset (e.g., protected area size)?
4. Is a paired test a two-sample test?
(Hint: No — it is a one-sample test on the differences.)
5. Why is the p-value not the only aspect that matters? Why does effect size matter as well?
6. You have heard that CBD production in plants might increase under mild drought stress, and *just for learning purposes*, you want to try it. Since you want to measure the concentration of that magical compound at different water availability levels, how would you design the experiments in terms of number of plants? Would you:
 - Apply a progressive drought treatment to the same plants and compare before and after measurements, or
 - Use two groups of plants, one well-watered and one exposed to drought?Would your choice of design affect the statistical test you use? Explain your reasoning and how you would decide on the experimental design.
7. After completing this exercise, I would contact the company to show them the results. Do you think this could be a good way to represent the results? What would you add/change?

If the image is not displayed [find it in this link:](https://unioxfordnexus-my.sharepoint.com/:i:/g/personal/zool2620_ox_ac_uk/IQCISLj_rsTmS6WAK4loaUgfAfmeNQsw07wSQfSKEgnAVig?e=sR7akP)

Change in key indicators after habitat protection

Each line represents one protected site. Dots show values before (●) and after (●) protection.

